WINDOWS | Identity, host and evidence

Windows PowerShell • replace <PLACEHOLDERS> • backtick continues a command; pipe/comma also continue • access/modules affect visibility

COMMAND / PATTERN	WHAT / WHEN / CHECK
01 IDENTITY / HOST / CONTEXT	

`$PSVersionTable.PSVersion`
`whoami /all`
`Get-CimInstance Win32_ComputerSystem`
	Shell, effective identity, domain/workgroup context.

`Get-CimInstance Win32_OperatingSystem |`
`  Select-Object Caption,Version,`
`  LastBootUpTime`
	OS/version/boot facts; do not infer OS from a port.

`Get-LocalUser` `Get-LocalGroupMember -Group "Administrators"`	LocalAccounts module; group names may be localised.

`Get-Location`
`Get-ChildItem -Force`
`Get-ChildItem Env:`
	Location, hidden items, env. Protect secret values.

02 FILES / ACCESS / HASH	

`Get-Content .\app.log -Tail 40`
`Select-String -Path .\app.log ``
`  -Pattern "fail|denied"`
	Tail and regex; retain original and surrounding context.

`Get-ChildItem C:\Lab -Recurse -File `` `  -Filter *.config`	Scoped search; access errors can mean missing evidence.
`icacls.exe C:\Lab\app.exe` `Get-Acl C:\Lab\app.exe | Format-List`	File and parent ACLs; writable alone is not escalation.
`Get-FileHash .\raw.txt -Algorithm SHA256` `Copy-Item .\raw.txt .\working.txt`	Copy can overwrite; hash source before working on it.

`Get-Content .\data.json -Raw |`
`ConvertFrom-Json`
`Import-Csv .\findings.csv`
	Parse into objects; inspect fields before filtering.

03 PROCESSES / SERVICES / TASKS	
`Get-Process | Sort-Object CPU -Descending |` `  Select-Object -First 8 Name,Id,CPU`	CPU is accumulated time, not live utilisation percent.

`Get-CimInstance Win32_Service |`
`  Select-Object Name,StartName,State,`
`  PathName`
	Service account, state and executable path.

`sc.exe qc <SERVICE>`
`Get-ScheduledTask |`
`  Select-Object TaskName,State,Actions`
	Check executable, permissions and actual trigger.

COMMAND / PATTERN	WHAT / WHEN / CHECK
04 NETWORK / RESOLUTION / REACHABILITY	
`Get-NetIPConfiguration` `Get-NetRoute -AddressFamily IPv4`	Addresses/gateway/routes establish next-hop context.

`Get-NetTCPConnection -State Listen |`
`  Select-Object LocalAddress,LocalPort,`
`  OwningProcess`
`Get-Process -Id <PID>`
	Listener -> PID -> process. Replace PID from output.

`Get-NetUDPEndpoint` `netstat.exe -ano`	UDP endpoints; native connection/PID fallback.
`Resolve-DnsName <HOST>` `Test-NetConnection <HOST> -Port 445`	Separate DNS and TCP tests; neither proves permission.

`Get-SmbShare`
`Get-SmbConnection`
`cmdkey.exe /list`
	Local shares, active SMB, stored targets (not passwords).

05 LOGS / FILTER / EXPORT	

`Get-WinEvent -FilterHashtable @{`
`  LogName="System"`
`  StartTime=(Get-Date).AddHours(-2)`
`} -MaxEvents 30`
	Provider-side time filter; bounded result count.

`Get-WinEvent -FilterHashtable @{`
`  LogName="Security"; Id=4625`
`} -MaxEvents 20`
	Failed logons; requires auditing/access. Add context.

`Get-Service |`
`  Where-Object Status -eq "Running" |`
`  Select-Object Name,Status |`
` Export-Csv .\services.csv ``
`  -NoTypeInformation`
	Objects -> select -> export; Format-Table is display only.

`Start-Transcript -Path .\session.txt ``
`  -NoClobber`
`Stop-Transcript`
	Start/stop session capture; protect secrets in output.

06 NEXT-DECISION CHECK	
`WHO RUNS IT? -> WHAT CAN I CHANGE?` `-> WHAT TRIGGERS IT? -> WHAT PROVES IMPACT?`	Correlate identity, writable input and execution trigger.
`Get-Help Get-WinEvent -Examples` `Get-Command *NetTCP*`	Discover local syntax; CMD cannot run these cmdlets.

Pathway: Security+ host/IAM/monitoring + CPSA/CRT/PEN-200 enumeration. Read effective context; preserve evidence; distinguish local, domain and application authority.